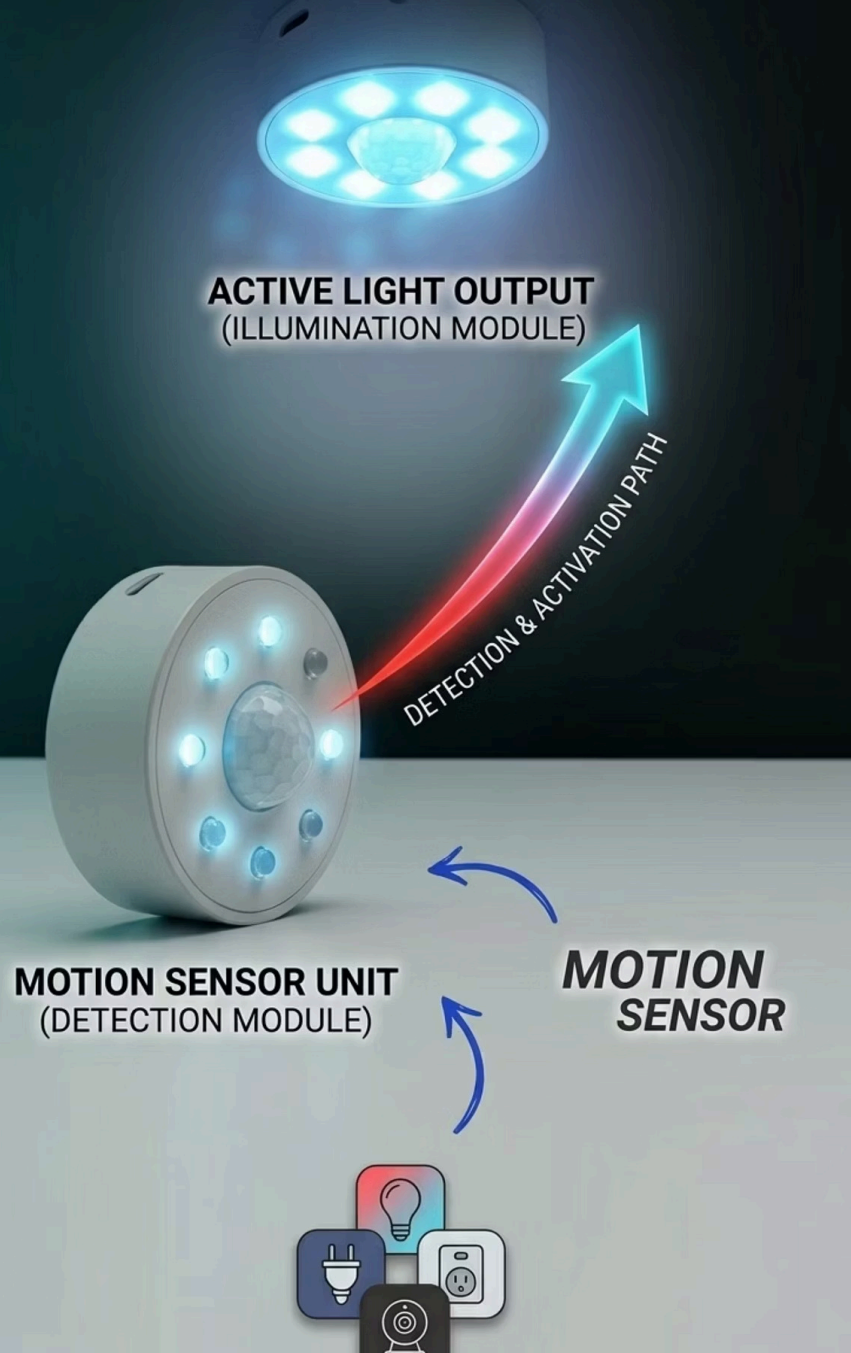




Smart Rechargeable Motion Sensor Light

With Type-C Charging & Automatic Night Detection

Presented by **Anushka Rani, Shruti Hirave & Shriya Pande**
TE — Electronics & Telecommunications

Introduction

Smart lighting systems represent a significant leap in energy-efficient technology. This project combines **passive infrared (PIR) motion sensing, ambient light detection,** and **USB Type-C rechargeable power** into a single, self-contained unit.

The Problem

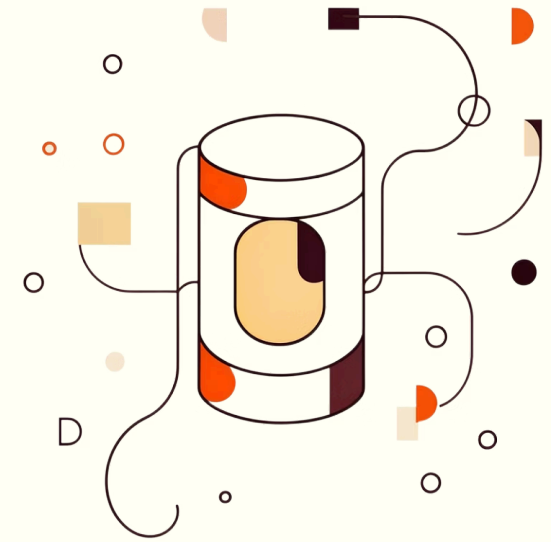
Conventional lights waste energy by staying on regardless of occupancy or ambient light levels.

Our Solution

A smart light that activates *only* when motion is detected *and* it is dark — saving power automatically.

Key Innovation

Rechargeable via Type-C — no disposable batteries, no hardwiring required.



Project Objectives

1 Automatic Motion-Triggered Illumination

Detect human presence using a PIR sensor and activate LEDs instantly, then auto-switch off after a set delay.

2 Night Detection & Daylight Override

Use an LDR to ensure the light operates *only* in low-light conditions, preventing unnecessary activation during daylight.

3 USB Type-C Rechargeable Operation

Integrate a lithium-ion battery with a Type-C charging module for convenient, cable-compatible recharging.

4 Energy Efficiency & Portability

Achieve low power consumption with a compact, wireless, wall-mountable design suitable for indoor and outdoor use.

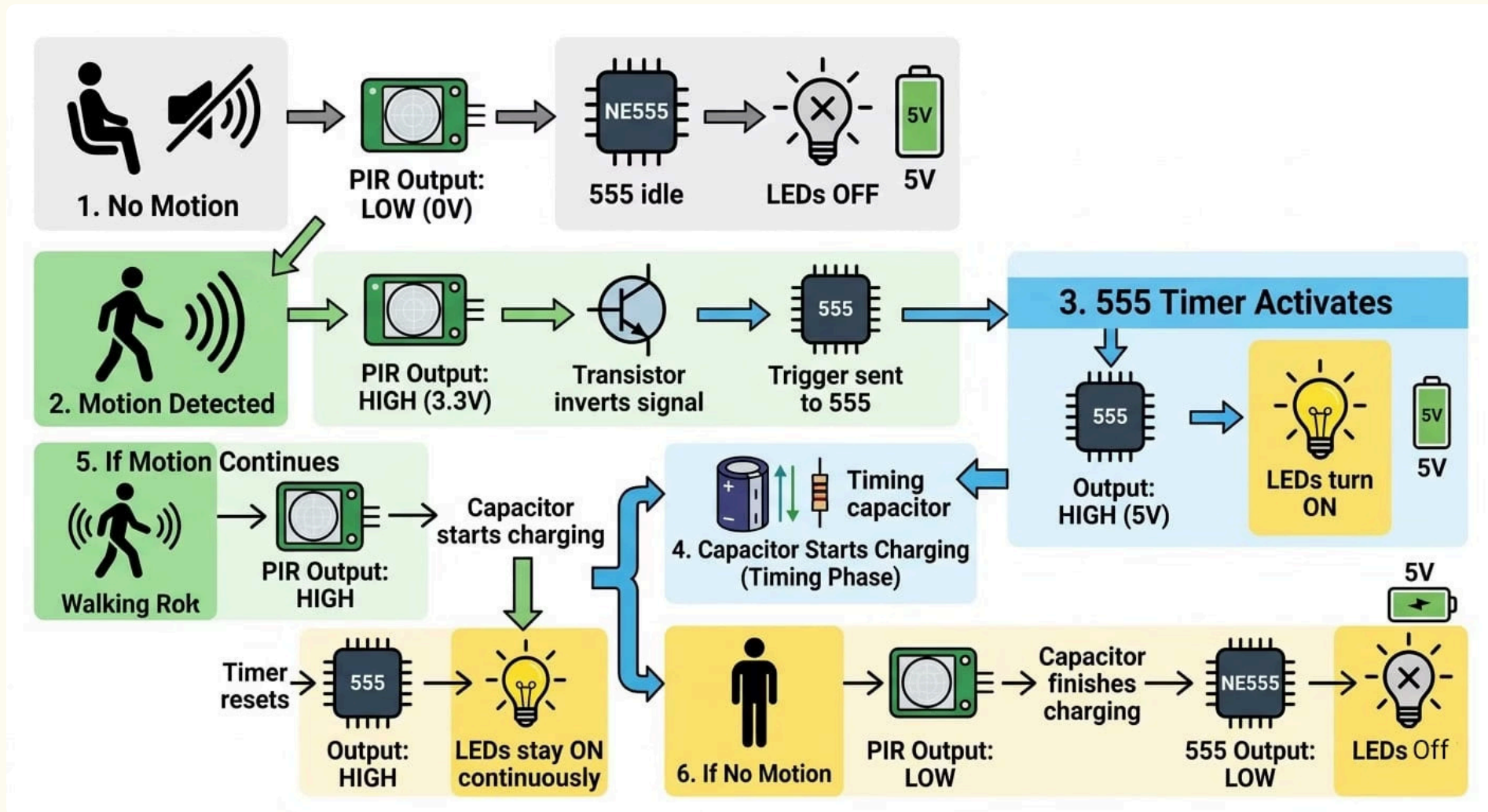
Components List

The following table outlines the key components used in the project along with their specifications and quantities.

Component	Specification	Quantity
PIR Motion Sensor	HC-SR501	1
LED Array	High-Brightness White LEDs	6
Lithium-Ion Battery	18650, 3.7 V, 2200 mAh	1
Type-C Charging Module	TP4056 with Protection	1
Boost Converter	MT3608	1
NPN Transistor	BC547	2
Resistors	100Ω, 47Ω, 1kΩ, 10kΩ, 22kΩ, 100kΩ, 220kΩ, 470kΩ, 1MΩ	as required.
Capacitors	47μF, 100nF, 100μF, 10μF	as required.
PCB	Custom PCB, ABS Plastic Housing	1
IC555	NE555	1
Connecting wires	PVC coated Tin plated wires	as required.
Filaments for 3D printing	plastic wires used for FDM/FFF printers	as required (probably 10 gm per design)

Block Diagram

The block diagram below illustrates the high-level architecture of the smart light system — from power input through sensing to LED output.



Input Stage

- Type-C charging port
- Li-Ion battery
- Boost converter (3.7 V → 5 V)

Sensing Stage

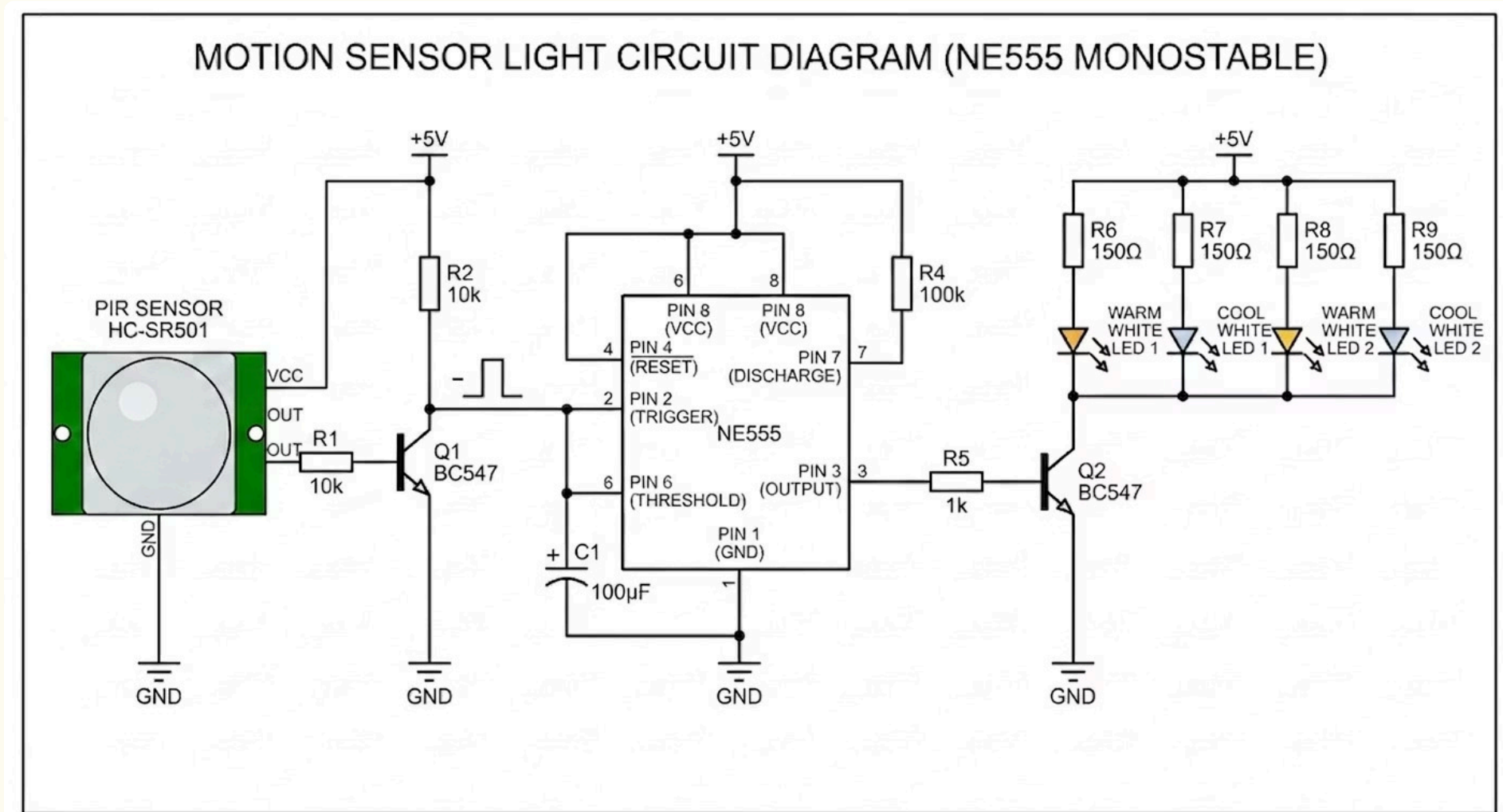
- PIR sensor (motion)
- IC555 (delay)
- logic.

Output Stage

- Transistor switching
- LED driver circuit
- Auto time-out control

Circuit Diagram

The circuit diagram provides a detailed view of all electrical connections between components. Refer to this schematic during assembly and troubleshooting.



Power Rail

5 V regulated supply from boost converter to microcontroller and LED array via transistor switch.

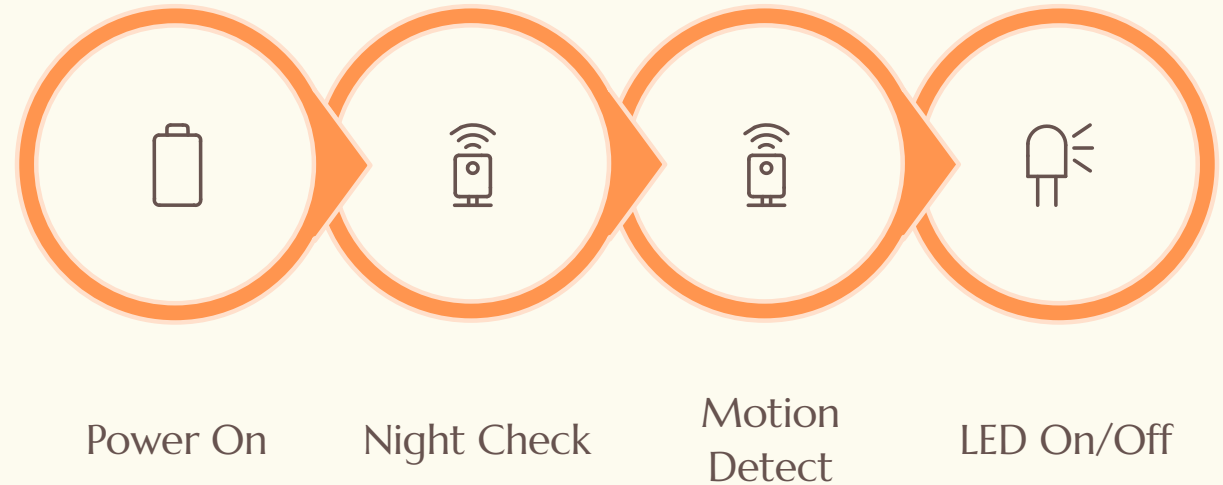
Sensor Interface

PIR output and LDR voltage divider both feed into digital/analogue input pins of the microcontroller.

Charging Circuit

TP4056 module manages Li-Ion charge/discharge with over-voltage and short-circuit protection built in.

Working Principle



The system continuously monitors ambient light. When darkness is detected, it arms the PIR sensor. On sensing motion, the microcontroller triggers the LED array via a transistor switch. After a pre-set delay (typically 30 seconds), LEDs switch off automatically — conserving battery until the next trigger event.

Project Progress Tracker

Track key milestones and deliverables for the project. Fill in dates as each phase is completed.

#	Date	Milestone / Task	Status
1	16 / 03 / 2026	Topic finalisation and team formation	✓ Completed
2	25 / 03 / 2026	Literature review and reference study	✓ Completed
3	09 / 04 / 2026	Component selection and procurement	✓ Completed
4	10 / 04 / 2026	Circuit design and simulation	✓ Completed
5	16 / 04 / 2026	Hardware assembly and soldering	✓ Completed
6	19 / 04 / 2026	Sensor calibration and testing	✓ Completed
7	27 / 04 / 2026	Final integration and performance evaluation	✓ Completed
8	28 / 04 / 2026	Report writing and presentation preparation	✓ Completed
9	10 / 05 / 2026	Presentation	✓ Completed

Cost Analysis

The table below provides an estimated cost breakdown based on standard market rates for each component. Update unit prices as per your actual procurement to calculate the final project cost.

Component	Qty	Unit Price (₹)	Total (₹)
PIR Motion Sensor (HC-SR501)	1	78	78
LED Array (High-Brightness White LEDs)	6	22	132
Lithium-Ion Battery (18650, 3.7V, 2200mAh)	1	115	115
Type-C Charging Module (TP4056)	1	22	22
Boost Converter (MT3608)	1	38	38
NPN Transistor (BC547)	2	3	6
Resistors	20	2	40
Capacitors	10	2	20
IC555 (NE555)	1	9	9
PCB + ABS Enclosure	1	150	150
Connecting wires	1 bundle	75	75
Filaments for 3D printing	1 bundle	250	250
Grand Total	—	—	₹ 1890

Conclusion

The **Smart Rechargeable Motion Sensor Light** successfully demonstrates a practical fusion of embedded sensing, power management, and user-centric design.



Energy Saving

Lights activate only when needed — motion + darkness — drastically reducing idle power consumption.



Rechargeable & Green

Type-C charging eliminates disposable batteries, making the device eco-friendly and cost-effective long-term.



Scalable Design

The modular circuit architecture allows future upgrades such as wireless control, solar charging, or IoT integration.

This project reinforces core principles of **Electronics & Telecommunications** — sensor interfacing, power electronics, and embedded logic — while delivering a tangible, real-world solution.

